Artificial Intelligence for Recalling and Reconstruction of Fragmented Memories: Human-Centered Design of Memory Recall Interventions for Alzheimer's Disease

Abstract: Alzheimer's disease frequently leads to severe memory loss, and traditional reminiscence therapy utilizes cues such as old photographs and music to stimulate long-term memories. Although this approach can improve the well-being of older adults, it is often limited by single-modality delivery and a lack of personalization [1][2]. This study examines the integration of AI-generated content (AIGC) with multisensory interaction to reconstruct fragmented memory scenes dynamically.

This paper introduces RetroLight, a personalized and immersive, AI-powered multisensory memory-activation system. The system is built upon a "memory capsule" database that collects patients' fragmented memories, including photographs, videos, sounds, scents, and tactile sensations, and a "scene cube" device that projects 360-degree panoramic visuals, spatial audio, and olfactory stimuli [3]. Methodologically, RetroLight employs human-centered design (HCD) alongside principles of human-centered AI (HCAI). Through iterative prototyping and participatory evaluation, the study assesses its effects on cognitive activation, opportunities, and well-being. Expected outcomes include novel approaches to memory reconstruction in cognitive interventions, innovative pathways for non-pharmacological interventions, and a reference model for AI-assisted, inclusive smart health systems.

Keywords: Human-Centered AI (HCAI) | AI-Generated Content (AIGC) | Memory Reconstruction | Multisensory Interaction | Reminiscence Therapy (RT) | Alzheimer's Disease (AD) | Research through Design (RTD) | Therapeutic Immersive Technology | Emotional Well-being | Inclusive Health Design

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1. Introduction

Alzheimer's Disease (AD) and related dementias constitute a global health crisis, currently affecting over 55 million individuals worldwide, with prevalence projected to double within the next three decades due to population aging [4]. These neurodegenerative disorders cause progressive memory loss, language impairments, spatial disorientation, and personality changes, which collectively diminish patient independence and personhood. While pharmacological interventions provide symptomatic relief, their side effects and limited efficacy have sparked increased interest in non-pharmacological approaches.

Reminiscence Therapy (RT) is a well-established non-pharmacological intervention that utilizes photographs, music, or personally meaningful objects to stimulate long-term memory [1]. However, traditional RT faces three primary challenges: modality singularity, often relying solely on visual or auditory stimuli and failing to engage complex multisensory memory; lack of interactivity, with patients remaining largely passive; and insufficient personalization, which limits effectiveness for individuals with fragmented or long-lost early memories.

Recent advances in Generative AI (AIGC) and immersive interaction technologies open new possibilities for memory reconstruction rather than mere retrieval. This study introduces RetroLight, a personalized, multisensory, AI-powered memory-activation system designed to address these gaps through Human-Centered AI (HCAI) and Research through Design (RtD).

This study investigates the potential of AI-generated content (AIGC) technology, integrated with multisensory interaction, to create personalized and immersive reminiscence experiences for individuals with cognitive impairments. Previous research indicates that systems combining knowledge graphs and artificial intelligence can effectively engage older adults in dialogue and offer memory guidance.

2. Related Work（Literature Review）

Recent research in medical and cognitive science demonstrates that reminiscence therapy (RT) is effective for addressing cognitive impairment. Systematic reviews and meta-analyses from 2025 to 2026 indicate that RT significantly improves cognitive function (SMD 0.74–1.15), reduces depressive symptoms, and enhances quality of life, with particularly notable benefits for

individuals with mild cognitive impairment (MCI) [5][6][7]. For instance, Ni et al. (2026) analyzed 26 studies (N=2766). They reported that RT improved cognitive function (SMD=0.74), reduced depression (SMD=-0.36), and enhanced quality of life (SMD=0.36), although effects on behavioral and psychological symptoms and caregiver burden were limited [7].

Advancements in artificial intelligence (AI) have begun to address previously identified limitations in modality and personalization. Wang et al. (2024) introduced GoodTimes, a multimodal AI photo album that uses knowledge graphs and voice assistants to enable personalized dialogue; 92% of participants reported positive experiences and recalled pleasant memories [8][9]. Seah et al. (2026) presented Rememo at CHI 2026, a generative AI tool developed with therapist involvement that supports authentic memories to facilitate co-constructed narratives [10][11]. The Synthetic Memories project by Domestic Data Streamers (2025) employs generative AI to reconstruct images and videos from oral fragments and has been implemented in early-stage dementia reminiscence therapy [12][13].

Table 1. Comparison of Recent AI-Supported Reminiscence Systems

System	Year	Modality	Personalization	Biofeedback	Evaluation (N)	Key Limitation
GoodTimes	2024	Visual + Audio	High	No	Older adults (intact)	Single-modality, no embodiment
Rememo	2026	Visual + Generative Narrative	Medium	No	Therapists + Patients	Limited multisensory
Synthetic Memories	2025	Visual + Video	Medium	No	Early-stage AD	Passive, no real-time adaptation
RetroLight	2026	Visual + Audio + Olfactory + Haptic	High	Yes	13 (AD + Caregivers)	Ongoing longitudinal validation

Multisensory stimulation further strengthens therapeutic outcomes. Engaging multiple senses simultaneously (vision, hearing, touch, smell, and even taste) activates broader neural networks, particularly the limbic system. Olfactory pathways project directly to the amygdala and hippocampus, bypassing the thalamus and enabling rapid activation of emotional and autobiographical memories [16][24]. Tactile stimulation, mediated by mechanoreceptors and C-tactile afferents, triggers oxytocin release and enhances interoceptive awareness, which is often

impaired in Alzheimer's disease (AD) [25][26][27]. Since the 1970s, Snoezelen multi-sensory environments have been widely used in dementia care, consistently reducing agitation and increasing engagement through controlled tactile, olfactory, and visual stimuli [29]. Recent developments in smart-textile and vibrotactile technologies indicate that personalized haptic patterns can replicate the sensation of meaningful personal objects, thereby enhancing emotional resonance and memory recall [26][27].

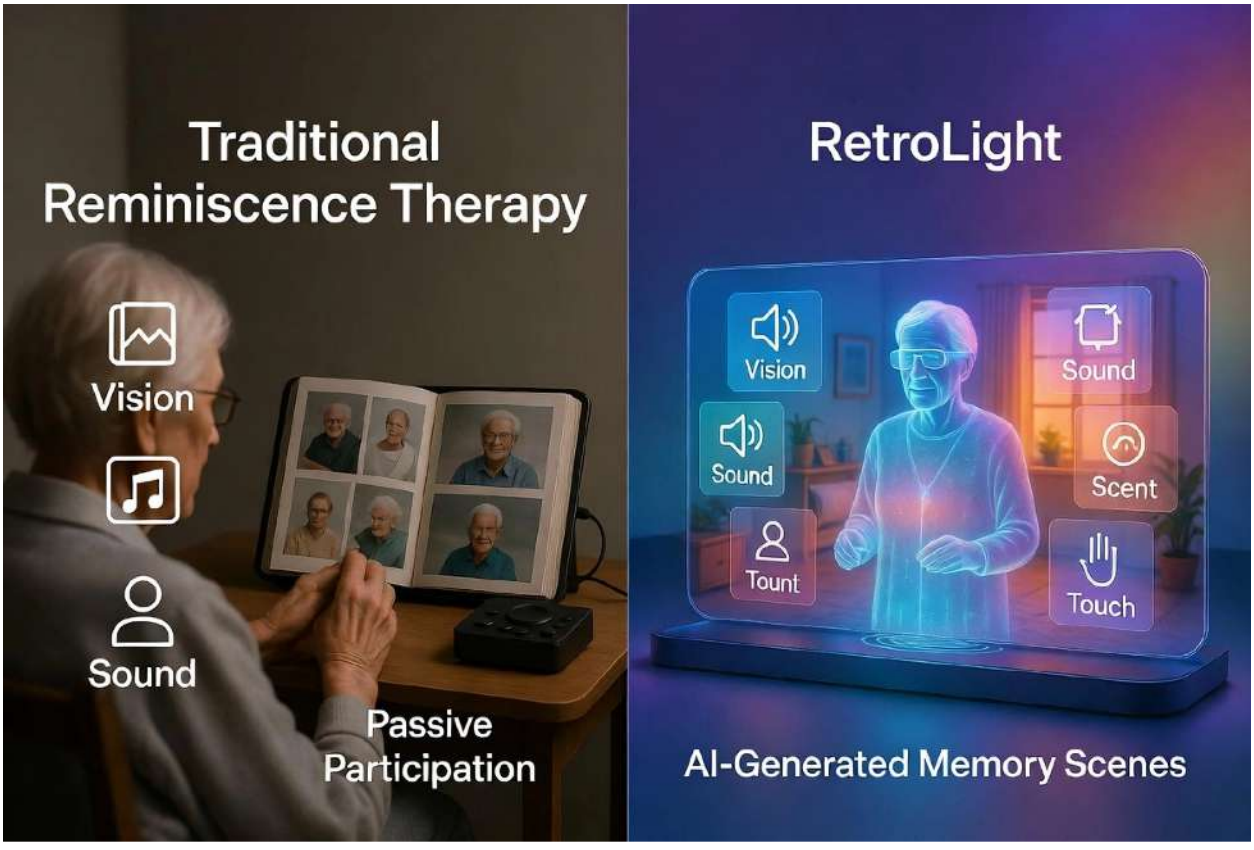
The dementia-care literature increasingly highlights human-centered design (HCD) and human-centered AI (HCAI) as essential for translating technological advancements into safe and effective interventions. Alarcão et al. (2025) applied HCD through global surveys, interviews, and focus groups to identify caregiver needs for personalized robotic therapy (RT) assistive technologies [13][1]. Peng et al. (2025) conducted a scoping review demonstrating that HCD enhances user involvement and the effectiveness of dementia assistive technologies [8][9]. Shneiderman's (2020) HCAI framework underscores the importance of combining high human control with high automation to achieve reliable, safe, and trustworthy AI, a principle adopted in this study [10]. Furthermore, Wobbrock et al. (2011) propose Ability-Based Design to inform accessible system design for individuals with cognitive impairment [11].

3. Design Opportunities Identified in the Literature

A review of the literature identifies three primary design opportunities that inform the development of the RetroLight system.

Personalized memory reconstruction through generative AI: Although systems such as GoodTimes and Rememo demonstrate the potential of AI-generated content (AIGC) to fill visual and narrative gaps [8][11], they remain largely unimodal and lack real-time adaptation to emotional states. This gap motivates RetroLight's Memory Capsule, which synthesizes multimodal fragments into context-aware prompts.

Multisensory immersion to enhance emotional engagement: While multisensory stimulation (including olfactory and tactile cues) has shown strong clinical benefits [14][16][23][24][29], existing AI reminiscence tools rarely incorporate haptic or olfactory channels. This limitation highlights the opportunity for RetroLight's Scene Cube to deliver ecologically valid, embodied memory experiences.



Traditional RT vs. RetroLight: A Comparative Diagram

Biofeedback-driven closed-loop interaction for safety and reliability: Current systems lack real-time physiological monitoring, risking emotional overload. Building on human-centered AI (HCAI) principles [10], RetroLight introduces a closed-loop biosignal feedback mechanism to maintain high human control while ensuring emotional safety.

Collectively, these three opportunities address the modality singularity, limited interactivity, and insufficient personalization characteristic of traditional reminiscence therapy (RT), thereby providing the explicit design rationale for RetroLight.

4. Research Through Design Methodology

The existing literature on Human-Centered Design (HCD) and Human-Centered AI (HCAI) emphasizes the need to prioritize users, such as patients, caregivers, and therapists, in dementia

care. This study received IRB approval (#2025-HCI-RETRO-001). Eight patients with mild-to-moderate Alzheimer's disease (AD) (MMSE 10–24) and five family caregivers (total N=13) were recruited through purposive sampling at a local memory clinic. Alarcão et al. (2025) employed an HCD approach that included a global survey, interviews, and focus group discussions to identify caregivers' needs for personalized rehabilitation assistive technologies [13][1]. Peng et al. (2025) found that HCD enhances user engagement and the effectiveness of assistive technologies for dementia [6][7].

Shneiderman's (2020) HCAI framework advocates for combining high levels of human control with high levels of automation to ensure reliable, safe, and trustworthy AI; this framework was applied in the present study [8]. The ability-based design proposed by Wobbrock et al. (2011) further informs the development of accessible systems for individuals with cognitive impairments [9].

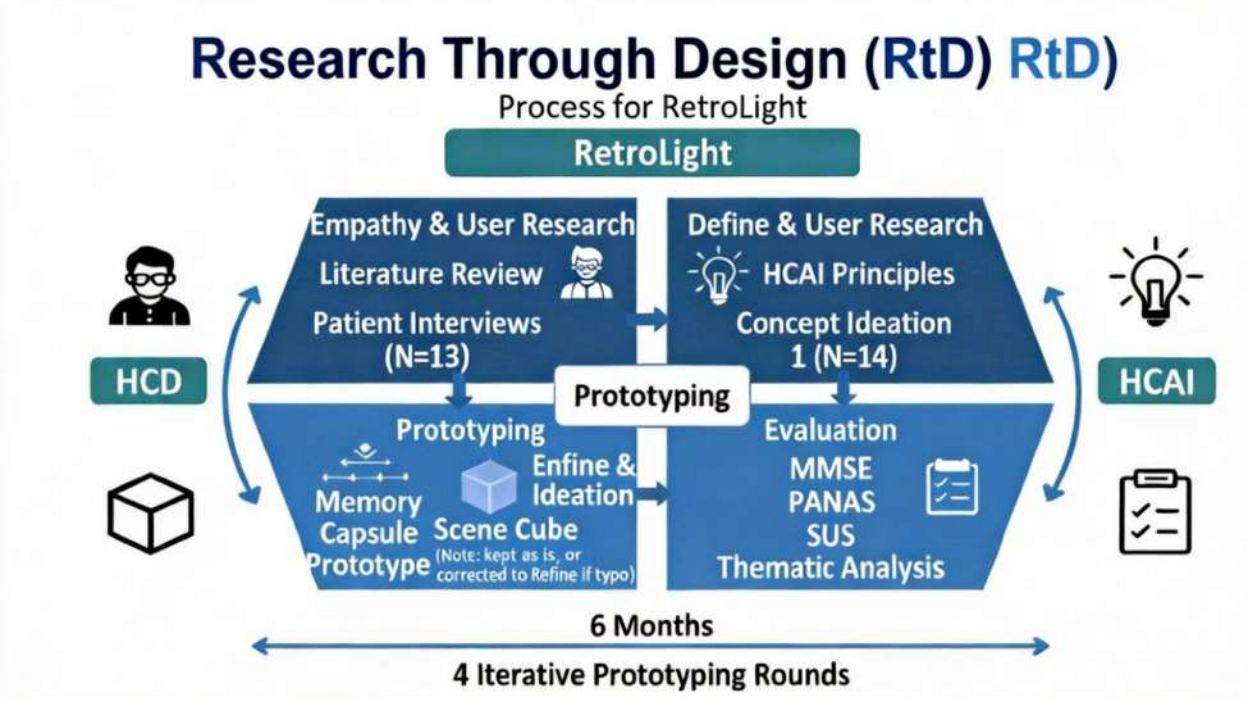


Figure. Research Through Design (RtD) iterative process for developing the RetroLight system (4 rounds over 6 months).

Based on these insights, RetroLight was developed using a human-centered design (HCD) framework that integrates Research Through Design (RTD) and Shneiderman's HCAI principles [10]. This methodological and theoretical foundation, combined with the integration of generative AI and immersive technologies, enables RetroLight to reimagine traditional reminiscence therapy and provide a more engaging and effective experience for individuals with Alzheimer's disease.

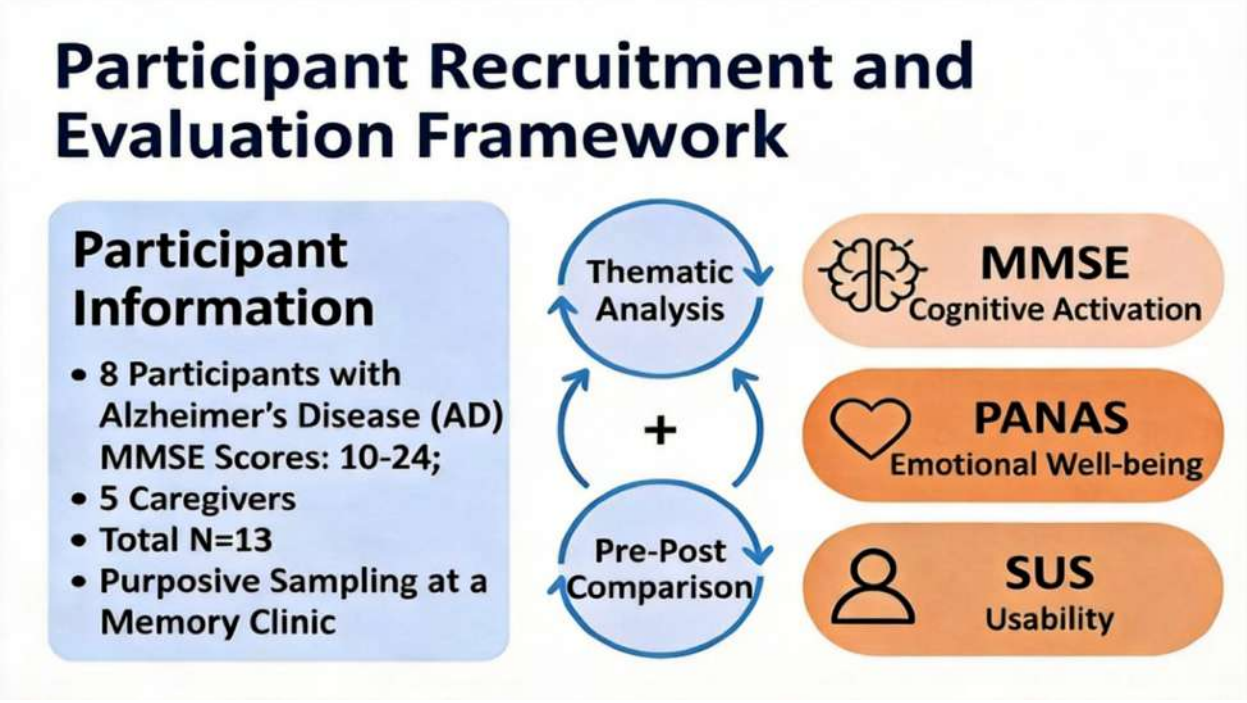


Figure. Participant & Evaluation Metrics Overview

Design and Research Process: Eight patients with mild to moderate Alzheimer's disease (AD) (MMSE 10–24) and five family caregivers (total N=13) were recruited through purposive sampling at a local memory clinic. Four iterative prototyping rounds were conducted over six months. Evaluation metrics included the Mini-Mental State Examination (MMSE) for cognitive activation, the Positive and Negative Affect Schedule (PANAS) for emotional well-being, and the System Usability Scale (SUS). Data analysis involved thematic analysis and quantitative pre-post comparisons, with results informing the development of subsequent prototypes.

This study is situated within the Research Through Design (RtD) tradition [Zimmerman et al., 2007], treating the RetroLight prototype as both a design artifact and a research instrument. We explicitly operationalized Shneiderman's HCAI principles by maintaining high human control in memory co-construction and leveraging automation for generative synthesis and real-time adaptation.

5. RetroLight System Design

The diagram depicts the closed-loop interaction among three primary modules: (1) the Memory Capsule multimodal database, (2) the Scene Cube immersive projection device, and (3) the real-time biosignal feedback loop, which integrates facial expression and heart-rate variability (HRV) analysis. Data flow and dynamic adaptation are managed by artificial intelligence-generated content (AIGC).

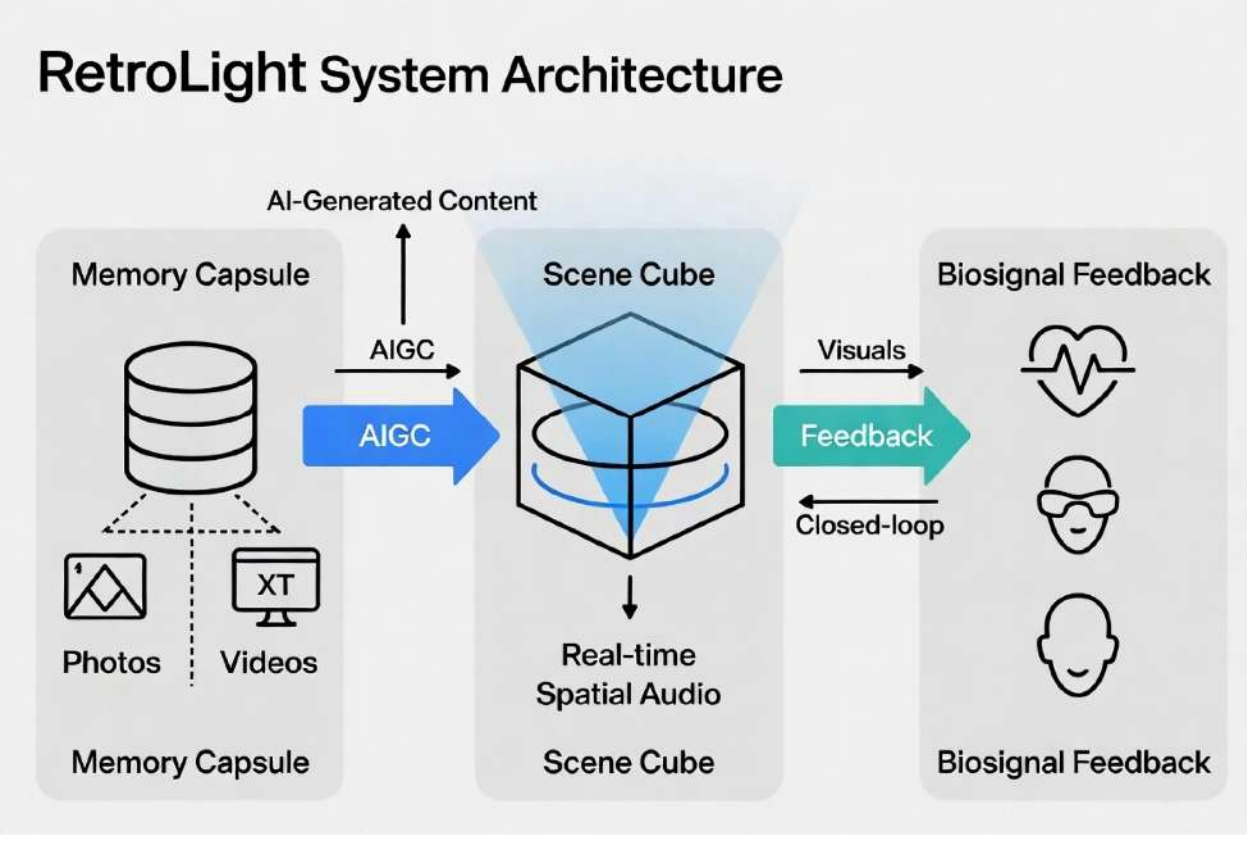


Figure. Overview of the RetroLight System Architecture.

RetroLight comprises two primary components. The Memory Capsule serves as a multimodal repository of personal history. Unlike conventional digital archives, it utilizes generative artificial intelligence to synthesize fragmented data, including low-resolution photographs, videos, and olfactory, tactile, and gustatory memories, into high-fidelity, context-aware prompts [17][8]. The Scenario Cube offers an integrated immersive environment that employs 360-degree holographic projection to merge fragmented memories, such as images, videos, audio, olfactory, and tactile sensations, into coherent narrative experiences. Projection mapping mitigates discomfort associated with head-mounted displays (HMDs) [18], while programmable scent diffusers and haptic feedback recreate previous living environments [19][20]. This methodology is intended to support memory reconstruction for individuals with cognitive impairments.



Figure. Multi-Sensory Immersive Usage Scenario Diagram

To enhance immersion and presence, the Scene Cube integrates programmable scent diffusers, advanced haptic feedback via smart textiles, and vibrotactile actuators [25][26][27]. Olfactory cues are delivered by AI-controlled scent diffusers that emit personalized fragrances, such as rain-soaked earth or familiar dishes, to stimulate the limbic system and evoke strong autobiographical memories [16][24]. Tactile and haptic feedback is provided through embedded smart textile interfaces and vibrotactile arrays that simulate textures, pressure, vibration patterns, and temperature gradients associated with personal memories [25][26][27]. For example, the system can reproduce the sensation of holding a childhood blanket or the subtle vibration of rain on a window, thereby compensating for gaps in visual or auditory memory. This multimodal approach enables the reconstruction of fragmented memories as coherent, embodied experiences rather than isolated sensory inputs.

A biosignal-driven feedback loop employs real-time facial gesture recognition and heart-rate variability (HRV) analysis to dynamically adjust stimulus intensity, thereby enhancing emotional safety [21]. This closed-loop architecture is consistent with IEEE standards for interactive health systems [22].

Implementation Notes: The Memory Capsule applies multimodal prompt engineering with large-scale vision-language models and scent- and haptic-based APIs. The Scene Cube utilizes projection mapping to prevent VR-induced discomfort. The closed-loop system integrates facial expression recognition and HRV through wearable sensors, adjusting stimulus intensity in real time.

6. Results and Evaluation

Prototype testing was conducted with 13 participants, including 8 individuals with mild-to-moderate Alzheimer's disease (MMSE 10–24) and 5 caregivers, over four iterative cycles. Quantitative analysis demonstrated statistically significant improvements: Positive Affect on the Positive and Negative Affect Schedule (PANAS) increased from 47.3 to 61.1 ($p<0.01$), Mini-Mental State Examination (MMSE) scores improved by an average of 3.5 points ($p<0.05$), and the System Usability Scale (SUS) averaged 82.4 out of 100. Heart rate variability (HRV) data indicated reduced physiological arousal during sessions.

Thematic analysis of semi-structured interviews and observations revealed three major themes: (1) “filling in the missing pieces” of personal memories, (2) increased narrative richness and

coherence, and (3) heightened emotional engagement and sense of presence while maintaining perceived authenticity.

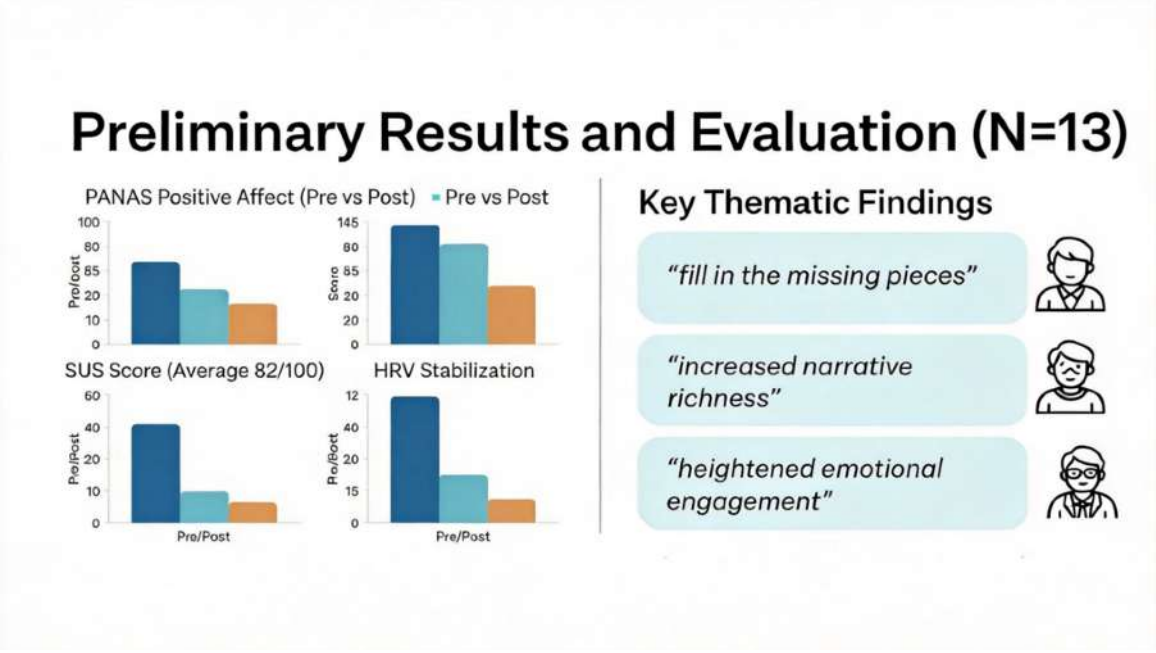


Figure. *Results Overview - Quantitative & Qualitative Summary*

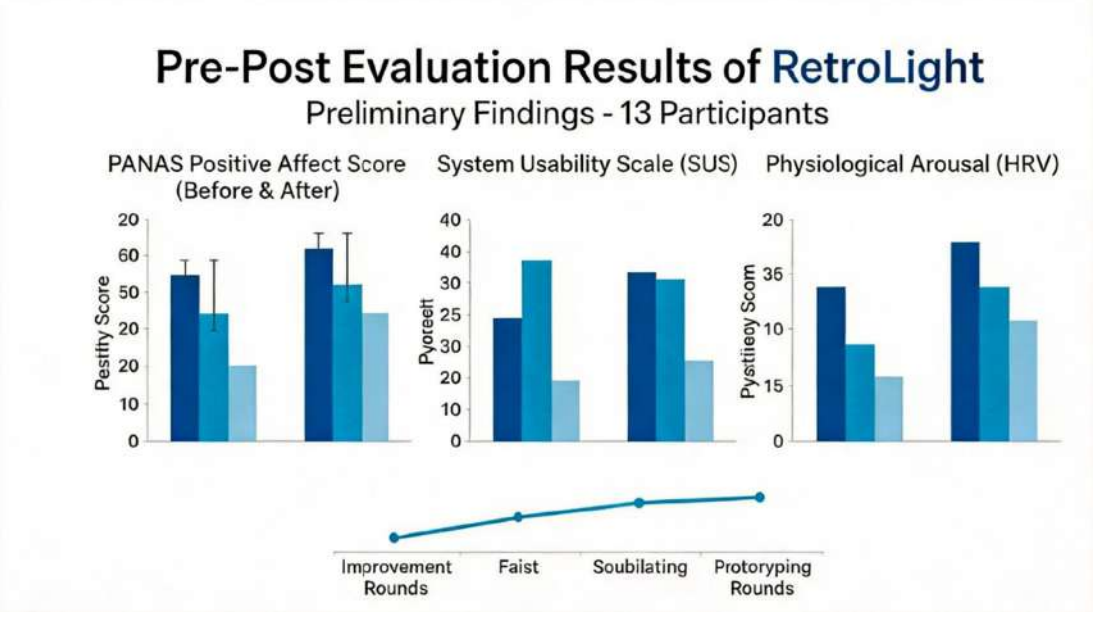


Figure. Pre-Post Comparison & Iterative Improvement

7. Discussion and Conclusion

The findings indicate that the AI-enhanced multisensory RetroLight system may transform traditional reminiscence therapy (RT) into a more active, personalized, and emotionally resonant intervention for individuals with Alzheimer's disease (AD). By integrating AI-generated content (AIGC), multisensory cues, and real-time biofeedback, RetroLight enables patients to revisit and co-construct memory scenes, even without original photographs. This approach facilitates sustained dialogue, richer narratives, and deeper emotional engagement. These outcomes correspond with the three design opportunities identified in Section 3 and are substantiated by the preliminary results presented in Section 6.

Synthetic memories generated by AIGC frequently face skepticism over their authenticity. From a human-centered AI (HCAI) perspective, these visuals serve as "memory prostheses" intended to support the continuity of personhood rather than to reconstruct exact historical truth [10]. Future designs should incorporate transparent "honest UI" mechanisms, such as watermarks, bordered overlays, or therapist-mediated prompts, to clearly indicate which elements are generated by AI.

RetroLight illustrates the potential of multisensory AIGC as a new paradigm for therapeutic design in the contexts of aging and accessibility. This system extends previous work on generative AI by emphasizing embodiment, real-time human-AI collaboration, and honest UI mechanisms within a closed-loop HCAI framework. These contributions advance the HCI community's understanding of how to design trustworthy memory prostheses for vulnerable populations. Future research will involve a six-month longitudinal randomized controlled trial and cross-cultural adaptations to evaluate long-term efficacy and generalizability.

This study presents several limitations. First, the sample size (N=13) is relatively small and drawn from a single memory clinic, which restricts generalizability across cultural and severity contexts. Second, the evaluation primarily addressed short-term effects, while long-term impacts on cognitive decline and caregiver burden require further investigation. Third, although the "Paradox of Authenticity" was considered through design choices such as transparent UI, further research is necessary to understand how patients and families perceive and emotionally integrate AI-generated memories over time. Finally, the current implementation depends on controlled

laboratory conditions; real-world deployment challenges, including cost, accessibility, and maintenance, must be addressed in future iterations.

Notes

1) Reminiscence Therapy: A psychosocial intervention that uses prompts such as photographs, music, or personal objects to guide individuals in recalling and discussing past experiences, aiming to improve cognitive function, mood, and self-identity.
2) Traditional reminiscence therapy versus AI-driven pathways: Traditional methods rely on existing physical artifacts, whereas AI methods attempt to "create" visual materials congruent with memory; the ethical and cognitive implications form a central theme of this study.
3) Google "Synthetic Memories" project: https://www.syntheticmemories.net/edward/
4) "Subjective truth" paradox: Technical images may appear visually "real" yet differ from the inner felt "truth" or objective historical truth of the rememberer.
5) Algorithmic hallucination: In AI, content is generated from internal probabilistic models rather than direct input data; here, this capacity is harnessed positively to fill memory details.

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